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STATISTICAL BIASES

TO WATCH FOR

FROM SAMPLING TO INFERENCE: WHERE RESULTS CAN DRIFT FROM THE TRUTH

Sampling	Measurement	Observation	Study design	Modeling	Inference
01 Undercoverage Part of the target population is missing from the sampling frame. "An email survey misses shop-floor staff without accounts."	02 Convenience sampling Easy-to-reach units differ from the population being estimated. "Day-shift coils stand in for all production."	03 Volunteer bias People who opt in differ on traits related to the result. "Only very dissatisfied customers answer the optional poll."	04 Nonresponse bias Nonrespondents differ from respondents on the outcome of interest. "High-scrap suppliers are less likely to return a quality survey."	05 Attrition bias Losses during follow-up change the groups being compared. "Units with rising vibration are lost before the six-month check."	
06 Survivorship bias Analysis includes only units that remain after earlier losses. "A plantwide coil rate based only on shipped coils omits upstream scrap."	07 Referral filter bias A referral path enriches the sample with unusual or severe cases. "A specialty lab's defect mix is used as the plant-wide rate."	08 Spectrum bias Test performance is studied on a case mix unlike actual use. "A detector tested on obvious pits and clean coils looks too accurate on borderline production."	09 Calibration offset A gauge reads consistently above or below a reference value. "A thickness gauge reads 0.02 mm high on every check."	10 Range-dependent bias A gauge's systematic error changes across its measurement range. "A load cell is correct at low load but high near capacity."	
11 Instrument drift A measurement system's response shifts over time. "A thermocouple reads progressively hotter between checks."	12 Hysteresis The same input reads differently on the upward and downward approach. "Pressure reads differently while rising than while falling."	13 End-digit preference Recorded values cluster at favoured digits or near thresholds. "Manual temperatures bunch at 0 and 5 despite finer resolution."	14 Misclassification Exposure or outcome units are assigned to wrong categories. "One mill calls borderline pits 'scratches' more often."	15 Insensitive measure A method misses meaningful differences it cannot resolve. "A coarse gauge hides small but important thickness changes."	
16 Verification bias Only selected units receive a definitive reference test. "Only camera-flagged coils receive destructive confirmation."	17 Incorporation bias The test under evaluation forms part of its own reference standard. "The camera's flag helps define the 'true' defect label."	18 Detection bias Comparison groups have outcomes sought or determined differently. "Trial coils receive closer visual inspection than controls."	19 Observer bias Knowledge or expectation shifts a subjective assessment. "An inspector expecting improvement calls borderline pits acceptable."	20 Recall bias Groups remember earlier exposures with different accuracy. "After a failure, crews recall unusual alarms more readily."	
21 Social desirability Respondents report what seems acceptable instead of what occurred. "A safety survey understates shortcut use."	22 Leading wording Question phrasing steers respondents toward an answer. "How much did the successful change help?" presumes a benefit."	23 Question-order effect Earlier questions change answers to later questions. "Asking about defects first changes later overall quality ratings."	24 Response-order effect The order of answer choices changes which are selected. "In a self-administered survey, the first listed cause is chosen more often."	25 Hawthorne effect People change behaviour because they know they are observed. "Operators follow checks more closely during an audit."	
26 Allocation bias How units enter comparison groups creates systematic differences. "Operators place easier batches on the new setting."	27 Chronological bias Groups enrolled at different times face different conditions. "A later trial benefits from a newer inspection system."	28 Historical-control bias A current group is compared with an earlier group under changed conditions. "A new rolling practice is compared with last year's product mix."	29 Lead-time bias Earlier detection lengthens observed time from diagnosis without delaying failure. "An early crack alarm seems to extend life, though failure time is unchanged."	30 Length-time bias Screen-detected cases look longer-lived because slow cases are easier to catch. "Periodic inspection catches slow cracks while rapid failures occur between checks."	
31 Immortal-time bias Pre-exposure event-free time is credited to the exposed group. "From casting, 'ever-reworked' coils look durable because they must survive to rework."	32 Starting-time bias Groups have different time origins, distorting event-time comparisons. "One line's failures count from casting, another's from shipment."	33 Adherence bias Those who follow a protocol differ from those who do not. "Comparing compliant shifts with all others rewards easier runs."	34 Regression to the mean Extreme baseline values can improve on repeat measurement and be mistaken for an intervention effect. "A fix gets credit when the worst scrap month naturally improves."	35 Confounding A common cause of exposure and outcome distorts their association. "Gauge mix drives both rolling speed and tensile properties."	
36 Confounding by indication The reason for an intervention also predicts the outcome. "Extra cooling targets difficult heats, so cooling appears harmful."	37 Collider bias Conditioning on a shared consequence creates a spurious association. "Among rejected coils, chemistry and cooling appear linked."	38 Overadjustment bias Controlling an intermediate variable removes part of a total effect. "Adjusting for cooling rate hides part of spray flow's effect on strength."	39 Functional-form bias An incorrect model shape systematically mispredicts parts of the range. "A straight line misses a U-shaped temperature-defect relation."	40 Regression dilution Error in one continuous predictor usually weakens a simple linear slope. "Noisy furnace temperatures mask a real strength relationship."	
41 Informative missingness Missing records are related to the unobserved values or outcome. "Sensors fail during the hottest runs, hiding extremes."	42 Ecological bias A group-level relation is wrongly applied to individual units. "Plant-level flow and yield are used to claim a coil-level effect."	43 Data leakage Holdout or future information enters model fitting or preprocessing. "A lab result measured later is used to predict an earlier outcome."	44 In-sample optimism Model performance measured on fitting data looks too favourable. "An AUC is reported on the coils used to tune the model."	45 Multiple-testing bias Many tests are tried but a nominal p-value is treated as isolated. "Fifty planned outcomes are tested; p = .03 is treated as decisive."	
46 Winner's curse The largest selected estimates tend to overstate the true effect. "The best of 100 alloy trials weakens on replication."	47 Data dredging Unplanned searches are presented as if they were prespecified tests. "Many subgroup cuts are tried, then the best is called the plan."	48 Selective result reporting Only favourable outcomes or analyses are disclosed. "Scrap, rework and yield are measured; only yield is reported."	49 Publication bias Studies with null or unfavourable findings are less available. "Only successful process trials enter the technical library."	50 Spin bias Interpretation suggests a stronger conclusion than results warrant. "A nonsignificant trial is headlined as 'proven effective.'"	